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Asymmetric Synthesis

Catalytic Asymmetric Aza-[2+2] Cyclization Reaction of Simple Ketoimine

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Abstract: Herein, we report the first catalytic enantioselective aza-[2+2] cyclization of simple ketoimines, thereby addressing a persistent challenge in four-membered aza-ring construction. While the aza-[2+2] cyclization represents the most straightforward approach to azetidine synthesis, previous methodologies have been strictly limited to ketoimines, all with an EWG at the central reactive carbon, and predominantly employing cyclic imine substrates. Through rational design of an in situ-generated magnesium catalytic system and systematic investigation of imine electronic effects, we have successfully developed an asymmetric protocol for simple ketoimines. This breakthrough enables efficient access to enantioenriched monocyclic azetidines with excellent stereocontrol. Moreover, a streamlined one-pot tandem oxidation readily converts these azetidines into valuable quaternary chiral β-lactams—privileged scaffolds that are prominent in numerous pharmaceuticals and bioactive agents. A combination of comparative studies, the calculation results of the HOMO and LUMO energies of different imines, relative control experiments, and a series of detailed NLE analysis revealed the coordination difference and the catalytic cycle. The catalytic protocol was used for the synthesis of a series of β-lactams containing aryl or alkyl groups, as well as for pharmaceutical active molecules' β-lactam-modifications. Importantly, the preliminary attempt to take advantage of the significant ring strain of β-lactam for peptide modifications was also achieved on tryptophan.

Introduction

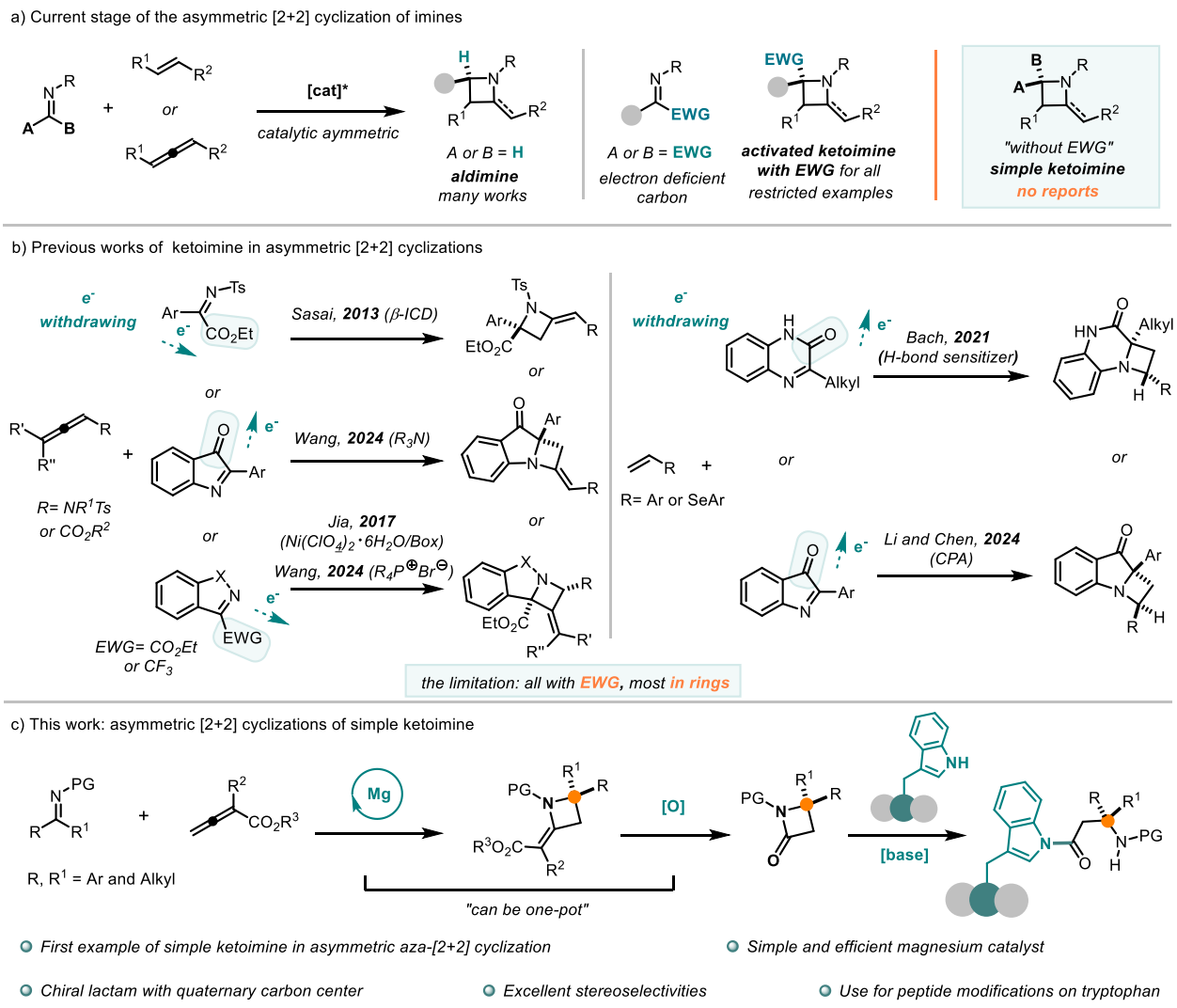
Azetidine and β-lactam scaffolds have garnered significant attention as privileged structural motifs in medicinal chemistry, with their importance in modern drug design continually growing.[1–6] Among various synthetic approaches, the asymmetric [2+2] cyclization of imines represents the most atom-economical and straightforward strategy for constructing these chiral, strained aza-heterocycles. Consequently, significant efforts have been devoted to developing catalytic [2+2] cyclization methods between imines and C=C bonds, such as alkenes or allenes. Over the past two decades, numerous elegant catalytic systems have been established for this transformation (Scheme 1a). Despite these advances, a fundamental limitation persists: while aldimines have been successfully employed to construct tertiary stereocenters,[7–12] the analogous asymmetric [2+2] cyclization of simple ketoimines remains unreported. Previous methodologies have been strictly limited to activated ketoimines (bearing electron-withdrawing groups at the central reactive carbon), leaving a significant gap in synthetic methodology. This limitation is particularly noteworthy because: (a) Simple ketoimines are more abundant and structurally diverse than their activated counterparts. (b) The resulting quaternary stereocenters in β-lactams are of exceptional value in pharmaceutical applications. In early work, the enantioselective organocatalyzed aza-MBH-type reaction of α-ketoimine esters and allenoates has been developed by the Sasai group for producing highly functionalized azetidines.[13] And the enantioselective [2+2] reactions of cyclic ketoimines adjacent to EWG have also been realized recently with the development of compatible catalytic methods independently.[14–17] In 2021, the visible light-mediated aza-Paternò-Büchi reaction[18–24] of quinoxalinones was established by Bach and Chien, through triplet energy transfer in a hydrogen-bonded complex using their classic chiral amide sensitizer.[25,26] Very recently, Chen and Li successfully described an elegant $B(C_6F_5)_3$/CPA-catalyzed asymmetric [2+2] cycloaddition of 3-indolone and aryl vinyl selenides, valuable chiral four-membered azetidines were delivered with excellent stereoselectivities (Scheme 1b).[27] Although these great achievements, the key issue is still not solved that all ketoimines applied in asymmetric [2+2] cyclization reactions need electron-withdrawing groups equipped to activate the C=N bond. One possible explanation for the established catalytic methods mainly relies on Lewis base activation of the electron-deficient alkenes

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Scheme 1. Current stage of the catalytic asymmetric [2+2] reactions between imines and a C=C bond.

or chiral acids for activation of ketoimines. Consequently, these systems inherently favor activated, electron-deficient ketoimines, leaving the more challenging but synthetically valuable simple ketoimines unexplored in asymmetric [2+2] cyclizations. To overcome this fundamental limitation, we proposed the employment of in situ magnesium catalysis[28–38] with modulated Lewis acidity could leverage the intrinsic N-nucleophilicity of imines,[39] which might be beneficial for the participation of electron-rich, simple ketoimines in the asymmetric aza-[2+2] cyclizations. Furthermore, we demonstrated the utility of this methodology by developing a streamlined protocol for β-lactam modification of tryptophan residues in peptides[40–50] (Scheme 1c), showcasing its potential in peptide engineering applications.

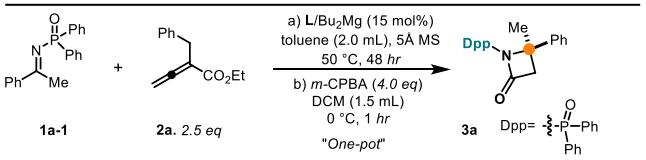
Results and Discussion

We commenced our investigation by examining the aza-[2+2] cyclization between ketoimine **1a-1** and allenoate **2a** using in situ-generated magnesium catalysts derived from BINOL ligands. Although small substituents at the C3-position of BINOL generally afforded high enantioselectivity, we observed that even subtle differences in ligand electronegativity profoundly impacted reaction efficiency (Table 1, entries 1–5). We attribute this to the modulated Lewis acidity of the resulting magnesium complexes, which significantly alters the reactivity of the coordinated imine. However, when ligands with larger substituted groups were used, the [2+2] reaction only gave very low chemical yields along with a decrease in enantioselectivities (Table 1, entries 6–8). After a general screening process, the *mono*-chlorinated BINOL **L3** led to the better cyclization efficiency with an excellent enantio-control result (81%, > 99% ee, Table 1, entry 3). Other ligands, such as those equipped with a hydroxy group or with stronger Lewis acidity endowed by more halogens on C6-positions, also gave unsatisfactory results (Table 1, entries 9 and 10). These results further prove that ligands with proper acidity would benefit the efficiency of the reaction, and small substitutions at the C3-position are good for the enantioselectivities. Besides the ligand, the variation of allenoates was also tried, and similar or

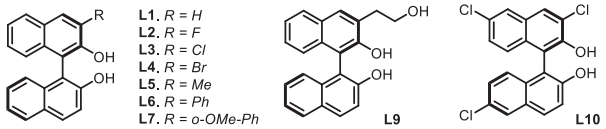
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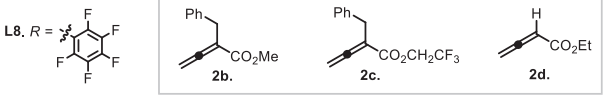
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Table 1: Reaction development.



Entries[a]	L	2	Yield of 3a	ee
1	**L1**	**2a**	38%	99%
2	**L2**	**2a**	26%	92%
3	**L3**	**2a**	81%	>99%
4	**L4**	**2a**	66%	98%
5	**L5**	**2a**	49%	96%
6	**L6**	**2a**	30%	88%
7	**L7**	**2a**	11%	53%
8	**L8**	**2a**	5%	34%
9	**L9**	**2a**	10%	29%
10	**L10**	**2a**	29%	98%
11	**L3**	with **2b**	72%	>99%
12	**L3**	with **2c**	80%	>99%
13	**L3**	with **2d**	29%	77%
14[b]	**L3**	**2a**	68%	>99%





[a] The reactions were carried out with imine **1a-1** (0.2 mmol) and allenoates **2** (0.5 mmol) under the magnesium catalyst (15 mol%) in the presence of 5 Å MS (100 mg), using toluene (2.0 mL) as the solvent. [b] The reaction was with 10 mol% catalyst and for 36 h.

decreased results were obtained accordingly (Table 1, entries 11–13). Further lowering the loading of the magnesium catalyst also decreased the reaction's efficiency (Table 1, entry 14).

Given the facts that the simple ketoimine can smoothly undergo direct [2+2] cyclization under the straightforward magnesium catalysis, we wondered about the rate comparison of imines with different electronic properties under the current catalytic strategy (Figure 1a). Thus, the aldimine **1a-2**, ketoimine **1a-1,** and keto ester imine **1a-3** were brought into the standard magnesium catalytic system, and it was observed that the initiation reaction rate of aldimine **1a-2** is slightly faster than that of ketoimine **1a-1**, and both of them are obviously faster than keto ester imine **1a-3,** which possesses the most electron-deficient carbon. The results significantly distinguish the current work from previous Lewis acids or Lewis base catalytic methods, which always use more activated ketoimines, equipping electron-withdrawing groups in the reactions. Frontier molecular orbital analysis within our comparative studies revealed that ketoimine **1a-1** possesses the highest HOMO energy (−8.199 eV) and a LUMO energy of −0.886 eV among the imines examined (Figure 1b). This electronic structure suggests that, for a mechanism dominated by the nucleophilicity of the imine nitrogen, **1a-1** should exhibit a faster reaction rate than the aldimine. Control experiments, however, demonstrated that **1a-1** reacts more slowly than aldimine **1a-2**. This apparent contradiction between the computational prediction and the experimental observation implies that additional factors are operative in the rate-determining step.

Then the systematic nonlinear studies including both of ketoimine **1a-1** and aldimine **1a-2** were carried out as a practical mechanism tool to probe catalytic complexes with the magnesium catalyst.[51–53] For ketoimine **1a-1**, it was observed stronger amplitude positive nonlinear effect ((+)-NLE) under higher catalyst loading amount (20 mol% compared with 15 or 10 mol%), while higher reaction temperature would slightly bring down the (+)-NLE intensity (Figure 2a, 1,2). These results suggest that higher temperature and lower catalyst loading amount are beneficial for reducing the (+)-NLE results. While the same conditions were applied for aldimine **1a-2**, the situation is obviously different. The variation range of nonlinear effects remains obvious with temperature changes (Figure 2a, 4), yet at higher temperatures and lower catalyst dosages, results approaching linearity are exhibited (Figure 2a, 3 and 6). Notably, the linear results are almost shown when **L4** with larger substitutions was used. However, under the same conditions, the nonlinear intensity of ketoimine remains very distinct. Further monitor of the chemical yields in the reaction of ketoimine **1a-1** showed higher ee values of ligand lead to higher conversions, especially at lower temperature, the reaction's efficiency is in consistent with the NLE data. These combinational results indicated that the cyclization reaction of aldimine has obviously lower amplitude of (+)-NLE, which might be contributed by the introduction of aldimine resulted in a higher speed for the depolymerization process of the inactive species.

Owing to the classic Kagan's model, a dominant formation of heterochiral complexes results in a (+)-NLE. Under such theoretical framework, the homo- and hetero-polymers containing two or more ligands and metals were supposed to be inactive catalyst that are formed in situ from the monoligated complex. The polymers would be depolymerized when Dpp-imines were introduced by the destructive coordination effects, and the depolymerized constant of aldimines $K_{DPa'}$ and $K_{DPb'}$ should be larger than K_{DPa} and K_{DPb} from ketoimines, which might lead to slower reaction rate of **1a-1**. For the possible mechanism of ketoimine **1a-1**, the tetradentate complex **I** might be generated, then the coordination of allenoate would release the coordination of the nitrogen atom of imine, the process might enhance the electronegativity while increase the electron density of the nitrogen atom in imine. Thus the whole coordination effects promoted the aza-[2+2] cyclization under the control of the magnesium complex, and the release of product and re-participation of imine **1a-1** regenerated the key mono-intermediate **I** (Figure 2b).

Next, a series of β-lactam with the chiral quaternary center was synthesized by the tandem cyclization and oxidation process under optimized conditions. Substituted phenyl groups with different electron properties were well

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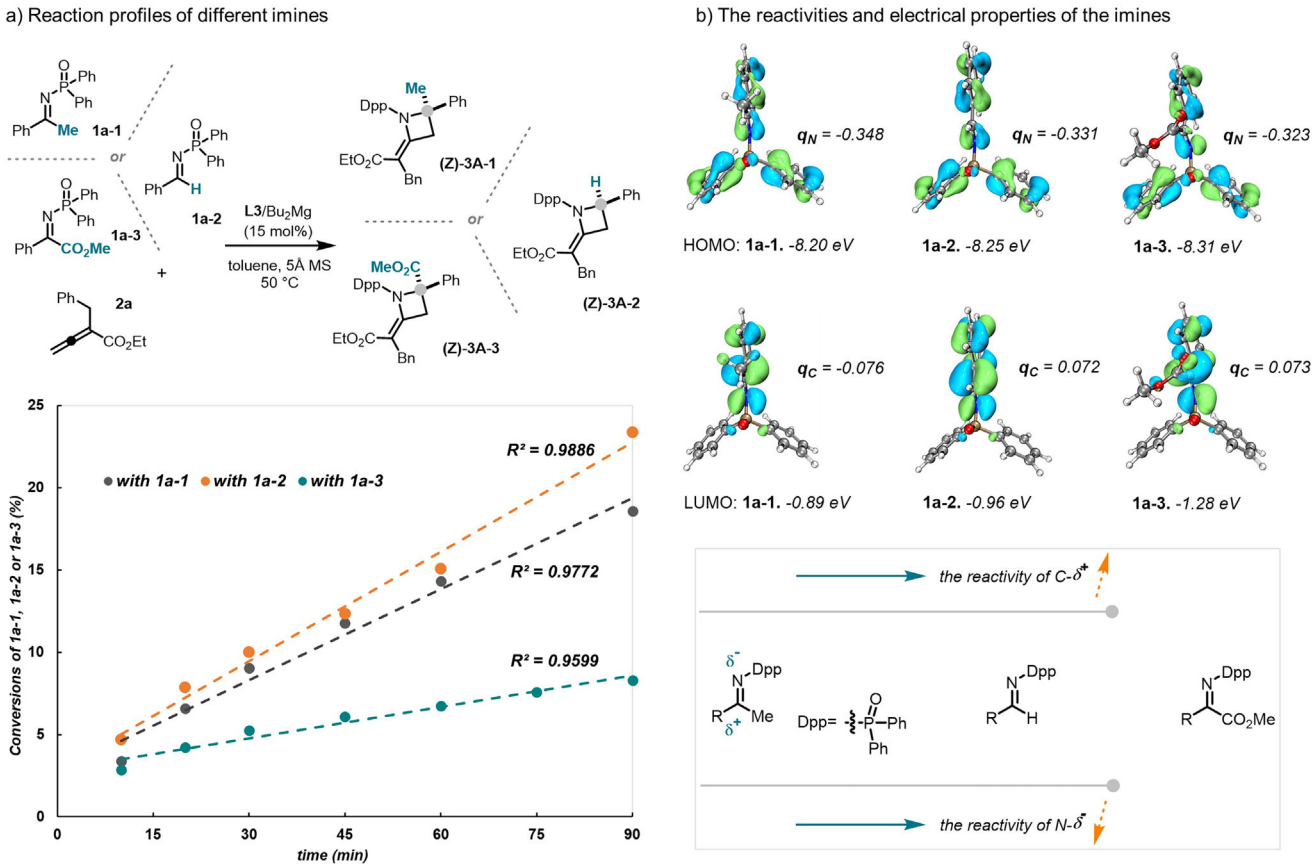


Figure 1. Comparison of the reactivity of imines with different electronic properties.

compatible in the catalytic system and provided the desired chiral β-lactams in high enantioselectivities with moderate to good chemical yields (Scheme 2). Some of the enantiomers were also smoothly obtained while the *ent*-**L3** was used. Naphthyl or heteroaryl groups are also applicable to the current magnesium catalyzed system. Multisubstituted aromatic ketoimines were synthesized and tolerable under the tandem synthetic protocol, leading the relative β-lactams with satisfied results. Finally, representative alkyl ketoimines were brought into the aza-[2+2] cyclization reaction, affording the desired full alkyl-substituted lactam adducts in high level of enantioselectivities (**3z, 3aa-3ad, 3ag, and 3ah**), and the absolute configuration of the β-lactam with the chiral quaternary center was determined by X-ray crystallization analysis of **3aa**.[54] More bulkier alkyl substituents were also tolerable, yet some led to low chemical yields (**3ae-3ah**). Moreover, some examples of the cyclization adducts without the oxidation step were also isolated with moderate yields and excellent enantioselectivities, thus the yields of most one-pot reactions should be mainly affected by the cyclization step. Notably, some heterocyclic structures were not transformed into lactams owing to the lower yields of the oxidation step (**(Z)-3N**, **(Z)-3O**).

To demonstrate the synthetic utility of the tandem reaction and given the facts that β-lactams are well-known privileged motifs in approved drug molecules, we used the one-pot cyclization and oxidation protocol for the β-lactam-editing of pharmaceutical-based compounds (Scheme 3). Under our in situ magnesium catalytic strategies, the compounds derived from β-estradiol, dapagliflozin, canagliflozin, etc, were successfully equipped with the β-lactam skeletons, and the quaternary chiral centers were well controlled by the configuration of the ligand selected in the modification process. Furthermore, we demonstrated the synthetic utility of this method by constructing drug-like molecules featuring non-natural β-amino acids or β-amino alcohols. For example, the carboxylic acid group of adapalene was utilized as the modification handle site for equipping with lactam, which can be smoothly transformed into relative amino acids **6** and amino alcohol **8**. And these compounds could be conjugated with a wide range of potential active fragments under mild conditions. Moreover, considering the extensive research interest of lithocholic acid within biomedical fields,[55,56] we also synthesized the β-lactam modified molecule **4g** in a scalable amount, and further transformed it into lithocholic amino acid **12** and alcohol **11**.

The obtained chiral quaternary β-lactam exhibited significant ring strain, and cleavage of the inner amide bond would naturally generate an β-amino acid residue. Thus, the structure of β-lactam might be an attractive manipulation tool for peptide direct modifications. Although small rings compounds were widely used as synthons in ring-opening reactions,[57,58] the strategy has rarely been applied for peptide or amino acids modifications.[59,60] Here, building upon our

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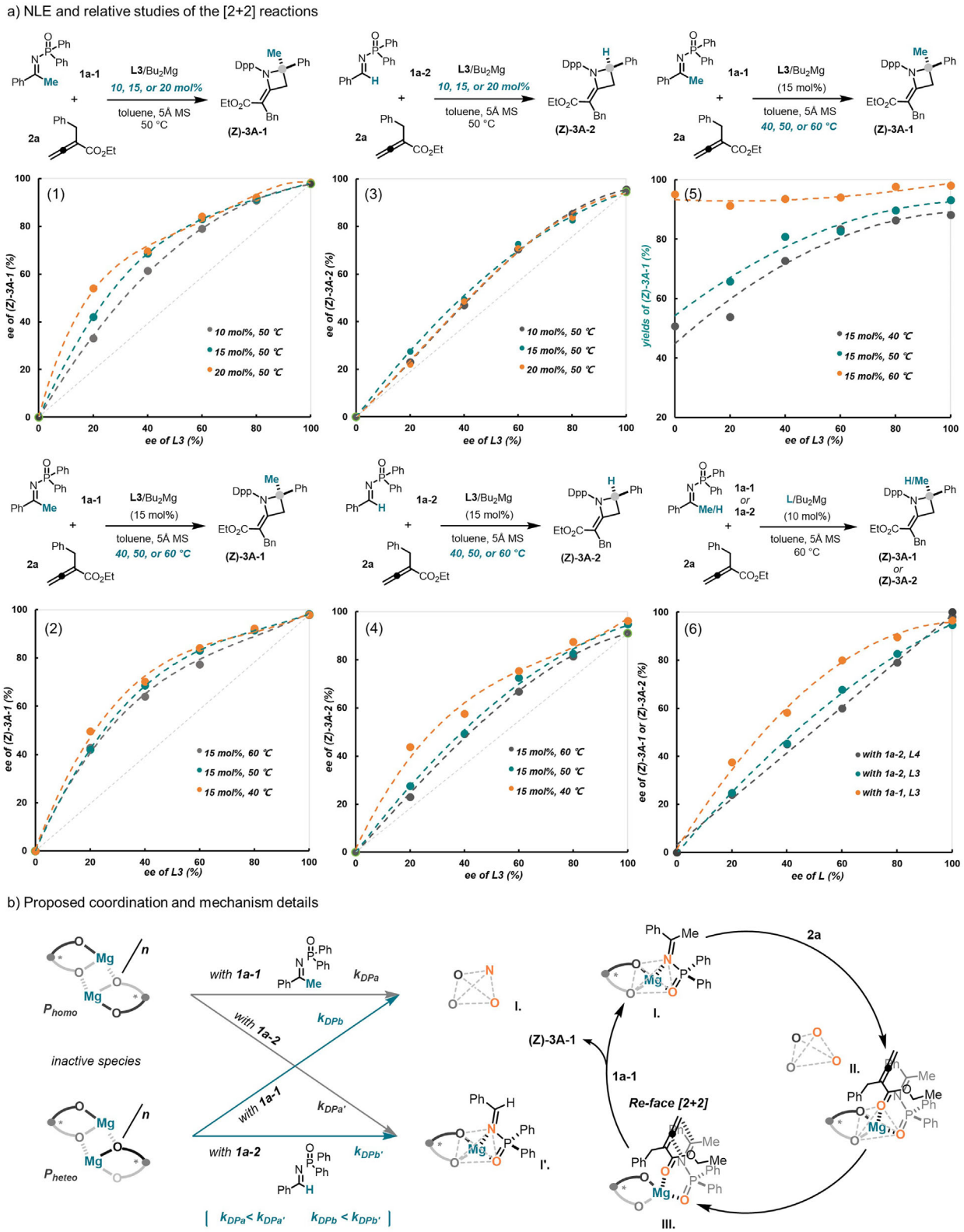
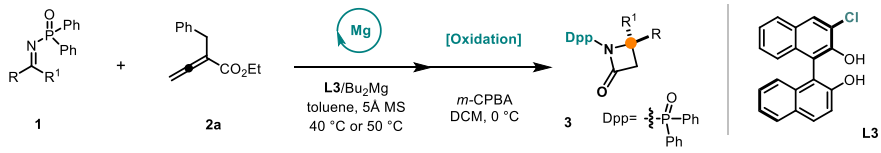


Figure 2. NLE relative studies, coordination details, and the catalytic cycle proposal.

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1 2a 3

3a. 81%, 98% ee

ent-**3a.** 75%, 99% ee[a)]

3b. 73%, >99% ee

3c. 61%, 98% ee

3d. 57%, 97% ee

3e. 71%, >99% ee

3f. 50%, 97% ee

3g. 58%, 98% ee

3h. 77%, >99% ee

3i. 57%, 97% ee

3j. 61%, 98% ee

3k. 69%, >99% ee

3l. 66%, 97% ee
ent-**3l.** 50%, 96% ee[a)]

3m. 73%, 99% ee
ent-**3m.**71%, >99% ee[a)]

3p. 33%, 96% ee

3q. 82%, 99% ee

3r. 61%, 99% ee
ent-**3r.** 63%, 98% ee[a)]

3s. 70%, >99% ee

3t. 41%, 97% ee

3u. 55%, 99% ee

3v. 57%, 98% ee

3w. 61%, 96% ee
ent-**3w.** 63%, 96% ee[a)]

3x. 66%, 98% ee
ent-**3x.** 63%, 98% ee[a)]

3y. 73%, 99% ee
ent-**3y**, 70%, >99% ee[a)]

3z. 43%, 84% ee

3aa. 48%, >99% ee

3ab. 62%, 95% ee
ent-**3ab.** 64%, 94% ee[a)]

3ac. 61%, 82% ee

3ad. 63%, >99% ee

3ae. 37%, 90% ee

3af. 16%, 87% ee

3ag. 42%, 83% ee

3ah. 40%, 76% ee

X-ray crystallographic analysis of ***3aa***

examples of the cyclization product:

(Z)-3E. 74%, >99% ee[b)]

(Z)-3N. 54%, 99% ee[b,c)]

(Z)-3O. 50%, 98% ee[b,c)]

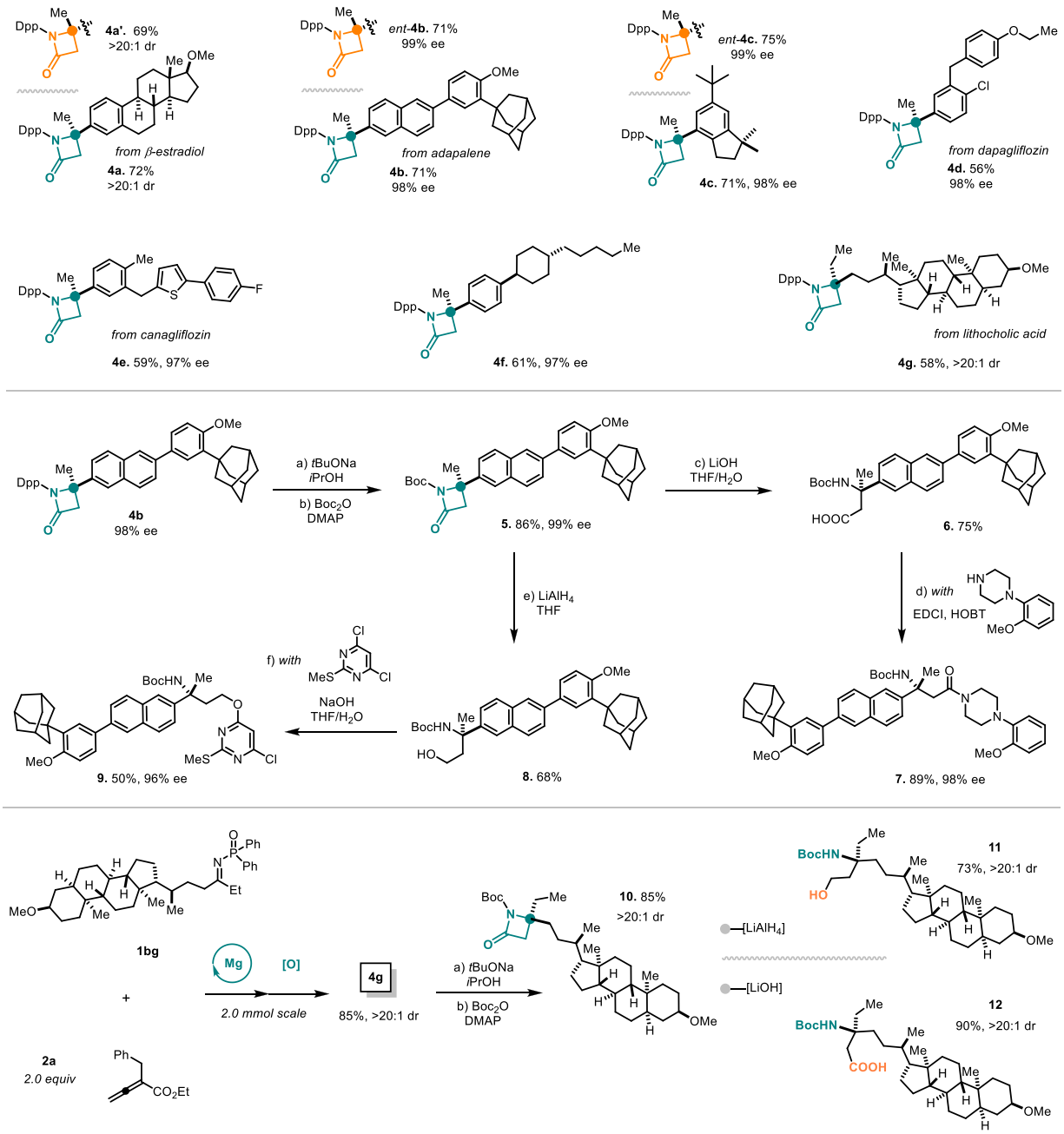
(Z)-3R. 68%, >99% ee[b)]

(Z)-3X. 70%, 98% ee[b)]

Scheme 2. Scope of the tandem aza-[2+2]-cyclization-oxidation for quaternary β-lactam synthesis. [a)]The *ent*-**L3** was used for the reactions. [b)]All Z:E ratio > 20:1. [c)]The reactions were not carried out in the tandem oxidation conditions for the lower yields of the step.

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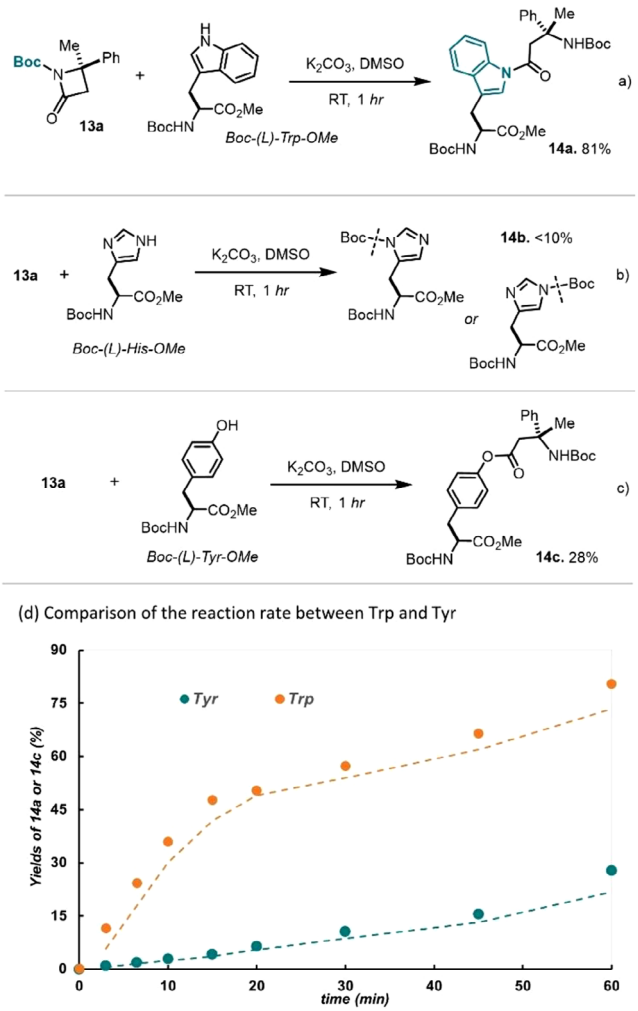
Scheme 3. *β*-Lactam editing of pharmaceutical active compounds and relative transformations.

synthetic platform, we gave a preliminary attempt to take chiral *β*-lactams into short peptide labeling studies. After the systematic screening process, we found the desired ring-opening reaction occurred at indole in tryptophan by simply treated with potassium carbonate in DMSO under mild conditions (Scheme 4a).[61] The structure of obtained adduct **14a** was progressively determined by *dept135* ^{13}C NMR and degradation experiment.[62] We further evaluated the transformation with other nucleophilic natural amino acids. While the phenolic hydroxyl group of tyrosine was also found to be reactive, affording the conjugated adduct **14c** (Scheme 4c), both the yield and reaction rate were considerably lower than those observed with tryptophan (Scheme 4d). Moreover, under the same conditions, the imidazole group of histidine was found to yield the corresponding Boc-protected adduct **14b** (Scheme 4b).

We subsequently applied these easily handled conditions to modify short peptides by incorporating pharmacologically active motifs using *β*-lactams bearing diverse protecting groups. This approach efficiently afforded corresponding peptides featuring an elongated backbone structure via tryptophan functionalization. A range of short peptides containing different amino acid residues was examined, successfully yielding the desired Trp-indole-N-elongated ring-opening adducts with yields ranging from 36% to 96% (Scheme 5, **15a-15o**). By varying the protection group, it was found that

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Scheme 4. Development of amino acid modifications with the chiral β-lactam. **13a** (0.15 mmol), amino acids (0.1 mmol), and K_2CO_3 (3.0 equiv) were used for these reactions. Other amino acids, including: Ac-(L)-Cys-OH; Boc-(L)-Ser-OMe; Boc-(L)-Lys-OH; Boc-(L)-Gln-OH, have no reactions in the presence of K_2CO_3, and for Boc-(L)-Arg-OH, only < 5% unidentified product was observed.

simple acyls such as benzyl or isopropanoyl can be tolerated under the alkaline conditions. Thus, we further drove the strategy to introduce pharmaceutical structures into the tryptophan-containing oligopeptides. Fragments coming from clofibric acid, (S)-ibuprofen, elaidic acid, etc, were modified as protected groups in β-lactams and smoothly introduced into the indole backbone in diverse short peptides (Scheme 5, **15r**-**15w**). Accordingly, the lithocholic acid-derived β-lactam was likewise employed for the site-specific modification of a peptide substrate under the same conditions, affording product **15x**, albeit in a lower yield. Furthermore, by incorporating an azide moiety into the acyl protecting group, we successfully conjugated representative fluorophores, including naphthalimide and pyrene derivatives, onto dipeptides via a CuI-catalyzed azide-alkyne cycloaddition reaction, providing the corresponding labeled products **19** and **20** in high yields.

Conclusion

In summary, we have accomplished a direct catalytic enantioselective aza-[2+2] cyclization of simple ketoimines using a concise in situ generated magnesium catalytic system. And a one-pot tandem oxidation with *m*-CPBA efficiently converts the resulting cycloadducts into β-lactams bearing chiral quaternary centers. Aryl- and alkyl substitutions are tolerable in the magnesium catalysis, thus the current method can be used as a tool for β-lactam-modification of pharmaceutical active compounds, as well as for the synthesis of relative drug-based amino acids, which might be potentially valuable non-natural synthetic motifs for peptides. Moreover, we successfully employed the obtained quaternary β-lactams for direct modification of tryptophan residues in short peptides, efficiently generating the anticipated indole-backbone-extended peptide sequences. This modification strategy further demonstrates potential as a versatile platform for site-specifically introducing pharmacologically active fragments into peptides. We believe this work represents a significant advance in catalytic asymmetric aza-[2+2] cyclization and the synthesis of quaternary β-lactams. Current efforts are focused on exploiting the coordination capability of the diphenylphosphoryl group to the in situ-generated magnesium center in other cycloaddition reactions, aiming to access additional valuable molecular scaffolds.

Acknowledgements

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Conflict of Interests

The authors declare no conflict of interest.

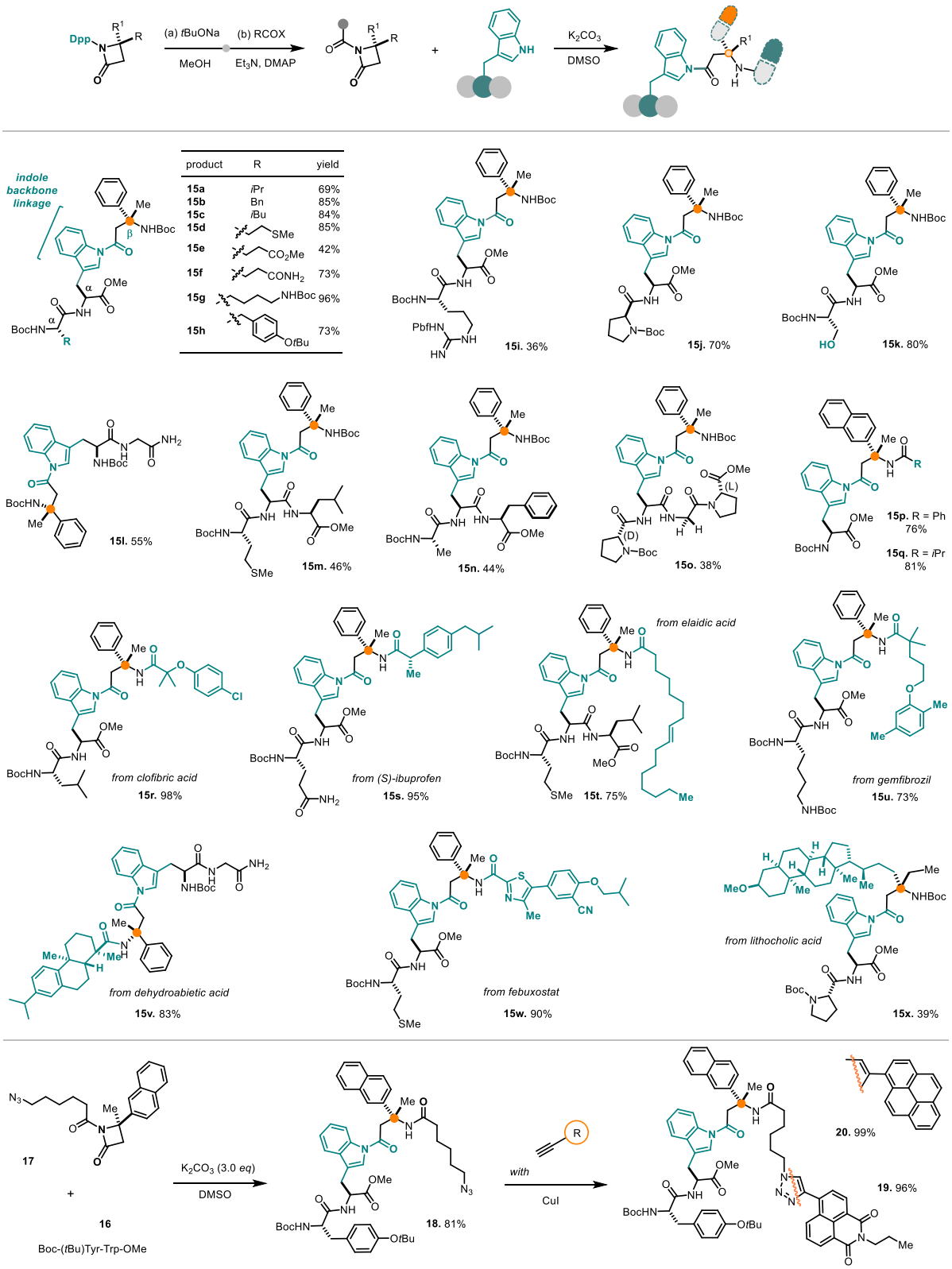
Data Availability Statement

The data that support the findings of this study are available in the Supporting Information of this article.

Keywords: Aza-[2+2]-cyclization · Magnesium catalysis · Peptide modifications · Quaternary β-lactam · Simple ketoimine

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Scheme 5. Scope of short peptides modifications on Trp.

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